AGRIVIVE / JHUM-NEST

MIZORAM / JHUM-CONTEXT / SLOPE MULCH RETAINING COLLAR

SERIES 17 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



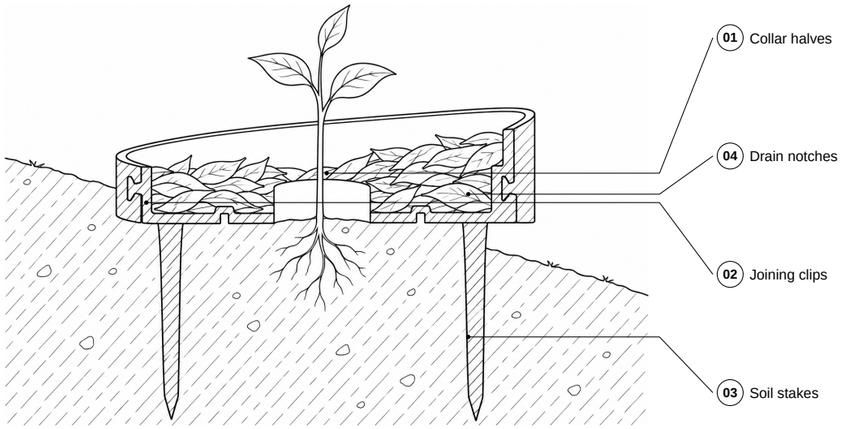
STATE CONTEXT / INDIGENOUS INSPIRATION

Jhum farming is part of Mizoram's agricultural setting; soil cover and runoff management depend on site conditions. JHUM-NEST is a new split mulch collar for trial plantings, rather than a claim about a traditional named implement.

Practice source: ICAR: Jhum cultivation and livelihoods in Mizoram | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 120 collar OD x 40 H

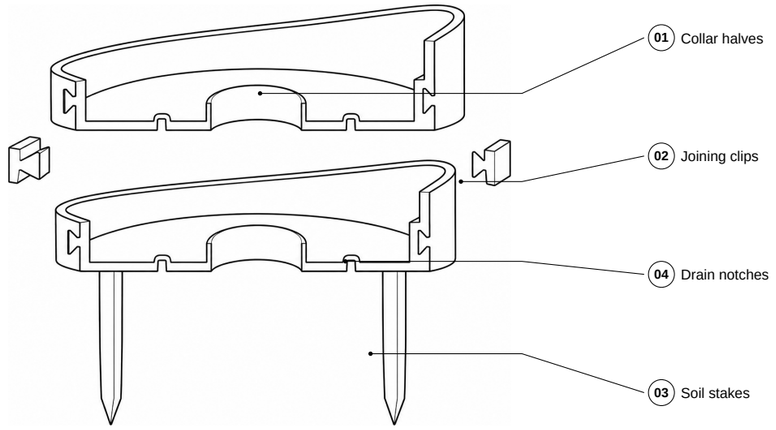


DESIGN / OPERATION NOTES

Join the halves around a seedling without contacting the stem.
Set stakes on the slope and place dry mulch in the outer ring.
Keep stem clearance and drainage notches unobstructed.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
120 collar OD x 40 H

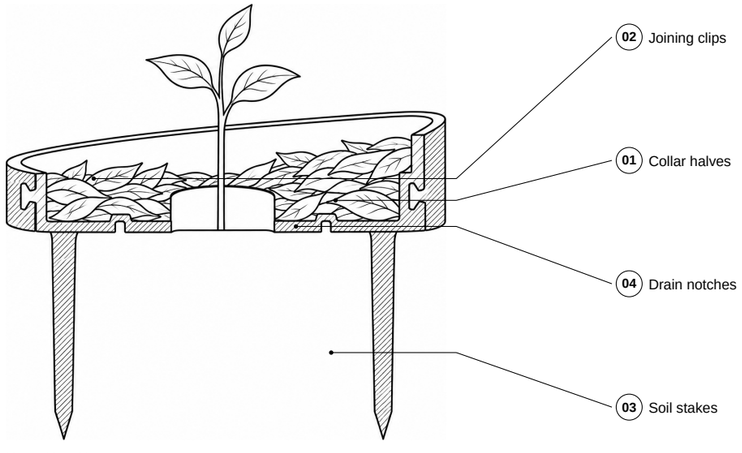


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

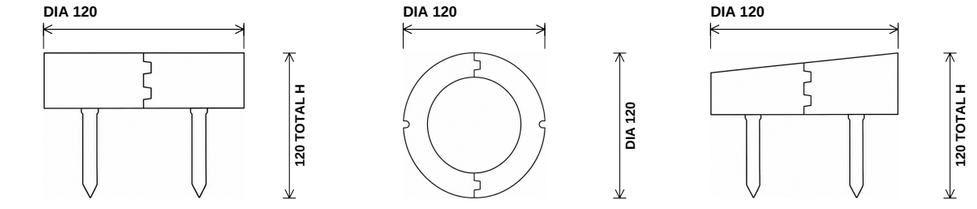
SECTION BASIS / mm
70 central opening ID; two 80 soil stakes additional



DESIGN / OPERATION NOTES

Set stakes on the slope and place dry mulch in the outer ring.
Keep stem clearance and drainage notches unobstructed.

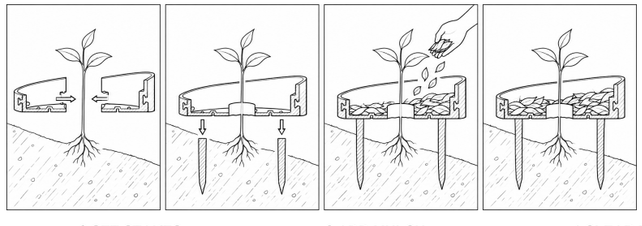
03 ORTHOGRAPHIC STUDIES / mm



FRONT / COLLAR **TOP / OPEN CENTRE** **SIDE / COLLAR**

Collar DIA 120 OD / DIA 70 opening x 40 maximum H; anchors extend 80 below collar.

05 INSTALLATION / OPERATING SEQUENCE



1 JOIN HALVES **2 SET STAKES** **3 ADD MULCH** **4 CLEAR DRAINS**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Collar OD / central ID	**120 / 70**
Maximum collar height	**40**
Stake extension	**80**

02 GEOMETRIC CHECK

DESIGN CHECK PENDING

TO BE TESTED

40 collar H + 80 stake = 120 total H
No irrigation storage; plant stem clearance required

03 PARTS / QTY

01 Collar halves	**2**
02 Joining clips	**2**
03 Soil stakes	**2 integral**
04 Drain notches	**Integral**

Collar/stakes/clips: candidate polymer. Biodegradation untested; mulch site-provided.

04 INTERFACES / CHECKS

Test slope retention, runoff bypass and stem clearance.

Check clip fatigue and soil anchorage.

Evaluate material durability; biodegradation is untested.

05 DATUMS / DRAWING BASIS

A: collar centre axis. B: collar base. Stakes extend 80 below B.

PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-17 / P01	JHUM-NEST	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	17 / 28